C.- Summary

Polymorphism:

The Three pillars of OO: encapsulation, inheritance (specialization) and polymorphism (abstraccion).

Meaning.

"many forms"

- Client code treats related objects as the same kind. and at runtime each one does the right thing.

Chess Example.

Before: each piece has its own method and the client uses instanceof with if/else. It works but it is hard to maintain, error-prone and verbose

Consequences.

Client code is shorter and cleaner, but you don't know exactly what it will do until runtime. One complexity is traded for another.

Extensibility

Improves, because new behavior goes into subclasses and the client stays unchanged.

In Java.-

- A chicken can be added to a list of FarmAnimal, but a FarmAnimal cannot be added to a list of chicken
- Overriding: a new definition of an inherited method
- Overloading: several definitions of a method in the same class.

Interface. Design.

Core idea: keep the simple simple. Design should start from people, not technology

Complexity: essential complexity belongs to the problem; accidentaly comes from our own choices.

For Laws of the humane interface (Raskin)

1. A computer must not harm your work or let it come to harm.
2. It must not waste your time or demand more work than necessary
3. It must respond to human needs and allow for human frailties
4. The user sets the pace of interaction

Interface approaches.

Letting the database, the programming language

or environment, or the plataform or framework drive the interface.

Abstraction.

What it is?

Bringing the details that matter in a context to the front and suppressing the rest. As an artifact, it is what a component exposses for others to use and rely on

Context - dependt. a Person class differs between a soccer tournament system and an airline reservation system.

Two Common problems.

- Leaky abstraction: other components rely on details that were never declared
- Over abstraction: too much is hidden, so userss lack control or access information.

The principel.

each component explicity and clearly declaress what it exposes no more and no less than others needed.